ORIGINAL RESEARCH

Actinidin as a Root Canal Irrigant: An *In Vitro* Investigation of Antimicrobial Activity Against *Enterococcus faecalis*

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Abstract

Aim: To evaluate the antibacterial efficacy of actinidin as a root canal irrigant against *Enterococcus faecalis*, compared with sodium hypochlorite and normal saline.

Materials and methods: An *in vitro* study was conducted on 33 extracted noncarious human premolar and molar teeth which was extracted for periodontal disease or orthodontic reasons. Specimens were inoculated with *E. faecalis* (ATCC strain) and incubated. They were randomly assigned into three groups (n = 11 each): group A: 5% sodium hypochlorite, group B: normal saline, group C: actinidin (0.5 gm mixed with saline). Bacterial viability was assessed using sterile paper-point sampling and colony-forming unit (CFU) counts obtained by plate-counting on MacConkey agar. CFUs were evaluated at 5 minutes and 48 hours using a digital colony counter. Statistical analysis was performed with Kruskal–Wallis H test and Dunn–Bonferroni *post hoc* comparisons.

Results: Group A showed complete bacterial elimination at 48 hours from 17.36 ± 4.98 (5 mins) to 0. Group C showed substantial CFU reduction from 20.90 ± 1.75 (5 minutes) to 1.63 ± 0.50 (48 hours). Group B showed the highest CFU values at both intervals (48.54 ± 34.32) at 5 mins and at 48 hours (20.72 ± 2.10).

Conclusion: Actinidin demonstrated significant antimicrobial activity against *E. faecalis*, suggesting its potential as a biocompatible adjunct or alternative to conventional irrigants in endodontic therapy.

Clinical significance: This study shows that actinidin, a natural enzymatic irrigant, provides significant antibacterial action against *Enterococcus faecalis*, outperforming normal saline and approaching the effectiveness of NaOCl. Its biocompatibility and ability to reduce bacterial load over time make it a promising adjunct or alternative in situations where NaOCl's toxicity is a concern. Actinidin may enhance root canal disinfection while minimizing tissue irritation, supporting safer and more patient-friendly endodontic care.

Keywords: Actinidin, Antimicrobial activity, Endodontics, *Enterococcus faecalis*, Irrigant, Sodium hypochlorite.

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Introduction

Root canal treatment (RCT) is a fundamental procedure in endodontics with the primary goal of eliminating infection from the root canal system to preserve natural teeth. The effectiveness of RCT is largely determined by the thorough elimination of microorganisms, especially those bacteria that form biofilms, from the intricately branched anatomical structures of the tooth.[1] One of the pathogens most frequently implicated in the continuation of endodontic infections is the gram-positive bacterium *Enterococcus faecalis*, which demonstrates remarkable resilience and survival potential even under the harsh conditions of root canal disinfection procedures.[2] Sodium hypochlorite (NaOCl) remains the gold standard irrigant for root canal treatments due to its broad antimicrobial activity against biofilm-forming microorganisms and its tissue-dissolving capability.[3] However, NaOCl has several limitations, including potential severe irritation to oral mucous membranes, unpleasant taste, and possible weakening of dentin when used in high concentrations or for extended periods.[4] These disadvantages have prompted the search for alternative irrigants that provide similar antimicrobial efficacy while being less toxic.

Actinidin, a cysteine protease enzyme derived from kiwi fruit (*Actinidia deliciosa*), has demonstrated antimicrobial activity in various applications, but its specific use in endodontics, particularly against *E. faecalis*, remains largely unexplored.[5] The rationale for investigating actinidin as a root canal irrigant is multifaceted. Enzymatic agents such as actinidin may provide a more biocompatible alternative to chemical agents, potentially reducing patient discomfort post-treatment.[6] Additionally, the proteolytic mechanism of action may offer enhanced effectiveness against biofilms where traditional irrigants may demonstrate limited efficacy. The investigation of actinidin's efficacy against *E. faecalis* is particularly relevant because this bacterium is prevalent in failed endodontic treatments and exhibits resistance to many conventional irrigants.[7] *E. faecalis* demonstrates the ability to invade dentinal tubules and form resistant biofilms, making it a challenging target for complete eradication.[2] Residual antimicrobial activity plays a critical role in preventing bacterial regrowth after instrumentation. Irrigants that provide both immediate antibacterial action and

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sustained residual effects can significantly influence endodontic treatment success. Therefore, evaluating irrigants not only for their initial bacterial reduction but also for their ability to prevent recolonization is essential in developing safer and more effective endodontic disinfection strategies. Thus, the aim of this study was to evaluate the antimicrobial efficacy of actinidin against *Enterococcus faecalis*, in comparison with sodium hypochlorite and normal saline.

Materials and Methods

This *in vitro* study was conducted at the Department of Microbiology at SRM Dental College, India, for a period of 3 months from June 2025 to August 2025. Based on the study by Agrawal et al., with a mean of 0.20 and standard deviation of ± 0.41, a sample size of 39 was calculated using G*Power software version 3.0 with 95% confidence interval and 5% margin of error.[8]

A total of 33 extracted human premolar and molar teeth were included in the study. Teeth indicated for extraction due to periodontal disease or orthodontic reasons were selected. Teeth with caries, fractures, resorption, open apices, previous endodontic treatment, and gross structural defects were excluded.

Specimen Preparation

Extracted teeth were debrided using a curette on external root surfaces, and surface disinfection was performed using 0.5% NaOCl for 24 hours. Access cavities were prepared, and canal instrumentation was performed using standard endodontic procedures to establish uniform canal dimensions. The working length was standardized, and canals were dried using sterile paper points.

Bacterial Strain and Inoculation

E. faecalis strain was procured from the Microbiology Department, SRM Dental College, Ramapuram, and a standardized bacterial suspension (1×10^8 CFU/mL) was prepared in nutrient-rich media for inoculating the specimens. Using a micropipette, 10 µL of the suspension was carefully injected into each instrumented canal. The specimens were then incubated at 37°C for 24 hours to allow bacterial colonization.

The 33 specimens were randomly assigned to three groups:

- Group A ($n = 11$): 5% sodium hypochlorite (Prime Dental Products Pvt. Ltd., India)
- Group B ($n = 11$): Normal saline (Baxter India Pvt. Ltd., India)
- Group C ($n = 11$): Actinidin (0.5 gm) (The Wholesaler Co. India) mixed with saline

Each canal was irrigated for 5 minutes with 5 mL of the respective solution using a 30-gauge side-vented needle.

Sample Collection for Colony-forming Unit (CFU) Assessment

To assess bacterial viability, sterile size 25 paper points were placed into each canal for 30 seconds to absorb canal contents. Paper points were then transferred into sterile Eppendorf tubes containing 1 mL saline.

CFU assessments were performed at two-time intervals:

- T1: 5 minutes after irrigation.
- T2: 48 hours after treatment, during which the samples were incubated to evaluate bacterial regrowth or residual antimicrobial activity.

Colony-forming Unit Quantification

Each sample was streaked onto three MacConkey agar plates using sterile cotton swabs to ensure uniform distribution. For every specimen, a separate set of plates was used to avoid cross-contamination. The plates were incubated at 37°C for 24 hours, and viable colonies were counted manually using a digital colony counter. For each specimen, CFU sampling was performed twice: T1 (5 minutes after irrigation) and T2 (after 48 hours). Separate sets of paper points and three independent MacConkey plates were used for each time interval per specimen to avoid cross-contamination. All specimen handling, plating, and CFU counting were performed independently by two calibrated investigators who were blinded to group allocation.

Statistical Analysis

Statistical analysis was performed using SPSS software version 25. Descriptive statistics were calculated for the mean distribution of data. Normality testing was conducted using the Shapiro-Wilk test. Due to non-normal data distribution, the nonparametric Kruskal–Wallis H test was employed, followed by Dunn–Bonferroni *post hoc* test for multiple comparisons. A p-value <0.05 was considered statistically significant.

Results

The mean CFU values for the three irrigant groups at 5 minutes and 48 hours are presented in Table 1 and Figs 1 to 3 . Mean values at 5 minutes: group A (NaOCl) 17.36 ± 4.98, group B (normal saline) 48.54 ± 34.32, and group C (actinidin) 20.90 ± 1.75.

At 5 minutes, group B exhibited the highest CFU counts, while groups A and C showed substantially lower values, indicating better immediate antibacterial activity. Mean values at 48 hours; group A (NaOCl): 0 CFU, group B (normal saline): 20.72 ± 2.10, and group C (Actinidin): 1.63 ± 0.50. At 48 hours, group A demonstrated complete bacterial elimination. Group C showed a marked reduction compared to its 5-minute value, while group B retained the highest bacterial load.

Table 1: Comparison of mean values for three irrigant groups at 5 minutes and 48 hours

						95% CI	
Groups		*N*	*Mean*	*Std. deviation*	*Std. error*	*Lower bound*	*Upper bound*
Group A	After 5 mins	11	17.36	4.98	1.50	14.01	20.71
	After 48 hours		0	0	0	0	0
Group B	After 5 mins	11	48.54	34.32	10.35	25.48	71.60
	After 48 hours		20.72	2.10	0.63	19.31	22.13
Group C	After 5 mins	11	20.90	1.75	0.53	19.72	22.09
	After 48 hours		1.63	0.50	0.15	1.29	1.97

Group-wise change between 5 minutes and 48 hours indicates that group A: CFU reduced from 17.36 to 0, indicating complete antibacterial efficacy, group B: CFU decreased from 48.54 to 20.72, showing minimal antimicrobial effect, and group C: CFU reduced significantly from 20.90 to 1.63, demonstrating strong time-dependent antibacterial activity.

According to the Kruskal–Wallis test (Table 2), statistically significant differences were observed among the three groups at both time intervals ($p < 0.001$). Dunn's *post hoc* test (Table 3) showed that at 5 minutes, group A differed significantly from group B ($p < 0.001$), while group C did not differ significantly from groups A or B. At 48 hours, all three groups showed significant differences from each other ($p \leq 0.018$), with group A showing complete elimination, followed by group C, and group B showing the least effect. Overall, sodium hypochlorite (group A) demonstrated the highest antibacterial efficacy at both time intervals. Actinidin (group C) showed significant antibacterial action, superior to saline and approaching the effectiveness of NaOCl at 48 hours. Normal saline (group B) consistently exhibited the least antibacterial effect.

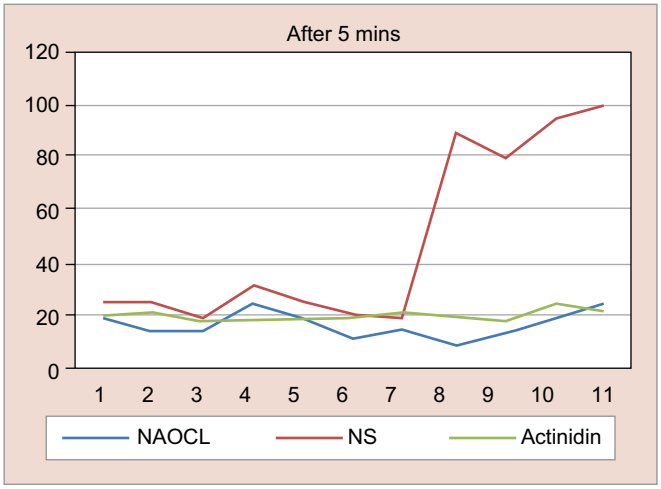


Fig. 1: Bacterial colonies estimated after irrigating within 5 minutes in all 11 samples of each group

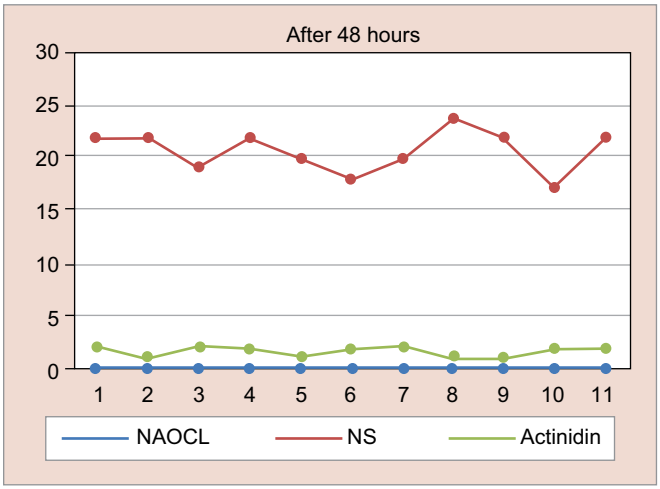


Fig. 2: Bacterial colonies estimated after irrigation in 48 hours in all 11 test samples

Discussion

This study compared the antibacterial efficacy of sodium hypochlorite (NaOCl), normal saline (NS), and actinidin against *E. faecalis*, a primary causative agent of chronic endodontic infections. The results demonstrated significant disparities among treatments in bacterial load reduction at both 5 minutes and 48 hours. Consistent

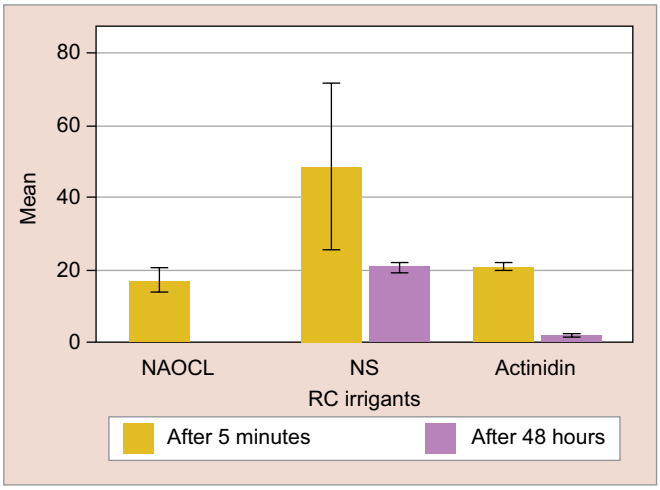


Fig. 3: Comparison between the effectiveness of three different irrigants within 5 minutes of irrigation and after 48 hours

Table 2: Comparison of CFU assessed after 5 mins and 48 hours of irrigating with three different materials

Variables	*N*	*H*	*df*	*p-value*
After 5 mins	33	14.292	2	<0.001*
After 48 hours	33	30.01	2	<0.000***

Kruskal–Wallis *H* test; *$p < 0.05$ is considered as statistically significant; ***$p < 0.001$ is considered highly statistically significant

Table 3: Dunn–Bonferroni multiple comparisons between bacterial colonies estimated at two levels among three irrigant groups

Variables	*Groups*	*H*	*SE*	*p-value*
After 5 mins	Group A–Group B	3.736	4.064	<.001***
	Group B–Group C	2.371	4.064	0.053
	Group C–Group A	1.364	4.064	0.517
After 48 hours	Group A–Group B	5.478	4.016	0.018*
	Group B–Group C	2.739	4.016	<0.000***
	Group C–Group A	2.739	4.016	0.018*

Dunn–Bonferroni corrections; *$p < 0.05$ is considered as statistically significant; ***$p < 0.001$ is considered as highly statistically significant

with previous investigations, normal saline exhibited the least antimicrobial effectiveness, showing the highest CFU values at both time intervals.[9] This finding confirms that saline possesses minimal antimicrobial activity and limited biofilm disruption capacity, rendering it inadequate as a sole irrigant for *E. faecalis* elimination from root canals. Statistical analysis confirmed NaOCl's superior efficacy compared to saline at both measured time points, validating its gold standard status in endodontic therapy.[1]

NaOCl achieved complete *E. faecalis* elimination within 48 hours, with zero detectable CFUs. This effectiveness is attributed to NaOCl's mechanism of compromising bacterial cell walls, denaturing proteins, and dissolving organic matter.[10] Similarly, Banu Jothi et al., stated that NaOCl showed CFUs of *E. faecalis* at 5 minutes, which were decreased to 0 after 48 hours, supporting the current study findings.[11] Also the current study findings reaffirm that NaOCl can effectively eradicate *E. faecalis*, making it a preferred choice in endodontic treatments. But also the utilization of effective aseptic techniques alongside a larger apical preparation size is also necessary.[12]

The current study validates previous findings demonstrating NaOCl's effective microbial reduction capabilities, though results may vary with different *in vitro* models and concentrations.[13] The proteolytic enzyme actinidin demonstrated noteworthy CFU reduction, performing better than saline but less effectively than NaOCl, particularly at the 48-hour time point.[14] Proteolytic irrigants, such as actinidin, possess enhanced ability to degrade biofilms and bacterial cell walls compared to inert irrigants.[15]

The statistical analysis revealed significant differences between actinidin and saline at 48 hours, supporting the potential role of enzymatic adjuncts, especially in clinical situations where NaOCl's corrosive activity may be contraindicated.[16] The Kruskal–Wallis H test and Dunn–Bonferroni *post hoc* analysis validated statistically significant activity differences among all three agents at both time points, specifically confirming NaOCl's overall superiority and actinidin's superiority over saline at 48 hours.[17] Due to *E. faecalis* resistance mechanisms involving robust biofilm formation and antimicrobial resistance, it remains a challenging target during root canal disinfection.[18] While NaOCl remains the optimal agent for comprehensive disinfection, actinidin may offer a novel adjunct or alternative in specific clinical circumstances. However, several aspects of actinidin's clinical application require further investigation, including optimal delivery methods, appropriate dosage, and potential synergistic interactions with other irrigants.[12] The *in vitro* system employed in this study cannot fully replicate *in vivo* conditions, including tissue structural complexities and immune system factors.[19]

Actinidin demonstrates promising antibacterial action and may serve as a biocompatible adjunct irrigant in clinical situations where sodium hypochlorite cannot be used or must be minimized due to tissue irritation or patient sensitivity. A study by Mugita et al. stated that actinidin reduced multispecies biofilm that was reconstructed in vitro, supporting the current study findings.[20]

The use of a standardized *E. faecalis* strain, controlled inoculation procedure, and two-time-point CFU evaluation strengthens the reliability of the findings. As an *in vitro* study, it does not replicate the full complexity of clinical environments, including dentinal tubule anatomy, host immune factors, and polymicrobial biofilms. The overall findings of this study demonstrated that sodium hypochlorite exhibited the highest antibacterial effect, followed by actinidin, while normal saline showed the least efficacy, consistent with the CFU reduction patterns observed. Further studies are required to evaluate the substantivity of actinidin, its interaction with other irrigants, its effects on dentin structure, and *in vivo* performance in clinical endodontic therapy.

Limitations of the Study

The study evaluated antimicrobial efficacy against a single microorganism (Enterococcus faecalis), whereas clinical infections are polymicrobial in nature. Additionally, the long-term substantivity, optimal concentration, and interaction of actinidin with other irrigants were not assessed.

Conclusion

Within the limitations of this *in vitro* study, actinidin demonstrated significant antibacterial activity against *E. faecalis* when used as a root canal irrigant. Although sodium hypochlorite remained the most effective agent, actinidin showed superior efficacy compared to normal saline, particularly at 48 hours. These findings suggest that enzymatic agents such as actinidin may serve as potential biocompatible adjuncts or alternatives to conventional irrigants in endodontic therapy. However, further research is required to optimize concentration, delivery methods, and substantivity, as well as to validate these results under *in vivo* conditions before clinical application can be recommended.

Clinical Significance

Actinidin demonstrated significant antibacterial activity against Enterococcus faecalis and showed better efficacy than normal saline, approaching the effectiveness of sodium hypochlorite at 48 hours. Its potential biocompatibility and enzymatic action suggest that it may serve as a safer adjunct or alternative irrigant in endodontic therapy, particularly in situations where sodium hypochlorite use is limited due to toxicity concerns.

Ethical Approval

This study was conducted in accordance with ethical standards and was approved by the Institutional Ethics Committee of SRM Dental College, Chennai (Approval No.: SRMU/M&HS/SRMDC/2026/UG/002).

Data Availability Declaration

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgment

Nil.

Authors' Contribution

TB: Conceptualization, methodology, investigation, data collection.
SR: Supervision, validation, review and editing.
BJA: Formal analysis, writing—original draft preparation.
LF: Literature review, data interpretation.
PD: Methodology support, investigation.
RM: Statistical analysis, review and editing.
DD: Data curation, visualization.
HR: Manuscript review, final approval.

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